



Misr

University for Science & Technology

Faculty of Engineering

ECE 342

COMMUNICATION I

Final Project

Universal Neural Demodulator: AI vs Conventional Performance Analysis

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I.Methodology

The study was conducted through MATLAB simulations where a message signal and carrier were defined to generate various analog modulation schemes including amplitude modulation, double sideband suppressed carrier, single sideband using the Hilbert transform, vestigial sideband through bandpass filtering, frequency modulation, and phase modulation. Each modulated signal was transmitted through an additive white Gaussian noise channel to evaluate performance under different noise conditions. Conventional demodulation techniques were applied, such as envelope detection for amplitude modulation, coherent detection for double sideband, single sideband, and vestigial sideband, and Hilbert-based extraction for frequency and phase modulation. The accuracy of recovery was assessed using mean squared error between the original and reconstructed signals, with error trends analyzed across a range of signal-to-noise ratios. In parallel, an artificial intelligence model was trained on a large dataset of noisy modulated signals with randomized message frequencies, using a fully connected neural network with nonlinear activation layers optimized by the Adam algorithm. The trained model was then employed to demodulate signals and its performance was compared directly with conventional methods, allowing a comprehensive evaluation of both traditional and AI-based approaches under identical channel conditions.

II.INTRODUCTION

Communication systems rely on modulation techniques to efficiently transmit information over noisy channels, making them fundamental to modern signal processing and telecommunications. Modulation allows a low-frequency message signal to be shifted to higher frequencies suitable for transmission, while demodulation is required to recover the original information at the receiver. Among the wide range of analog modulation schemes, amplitude modulation, double sideband suppressed carrier, single sideband, vestigial sideband, frequency modulation, and phase modulation each offer distinct trade-offs in terms of bandwidth efficiency, robustness to noise, and implementation complexity. Traditional demodulation methods, such as envelope detection and coherent detection, have been widely used for decades, but they often suffer performance degradation in low signal-to-noise ratio environments due to their reliance on precise synchronization and filtering.

With the increasing demand for reliable communication in noisy and unpredictable environments, artificial intelligence has emerged as a promising alternative to conventional demodulation. Neural networks, in particular, can learn complex nonlinear mappings between noisy modulated signals and their original message content, offering adaptive recovery without explicit carrier synchronization or filter design. This study explores both conventional and AI-based demodulation approaches through MATLAB simulations, where multiple modulation schemes are implemented, transmitted through additive white Gaussian noise channels, and evaluated using mean squared error as a performance metric. By comparing traditional techniques with a trained neural network across varying noise levels, the work highlights the strengths and limitations of each

approach and demonstrates the potential of AI to enhance communication system reliability.

III.Theoretical equations

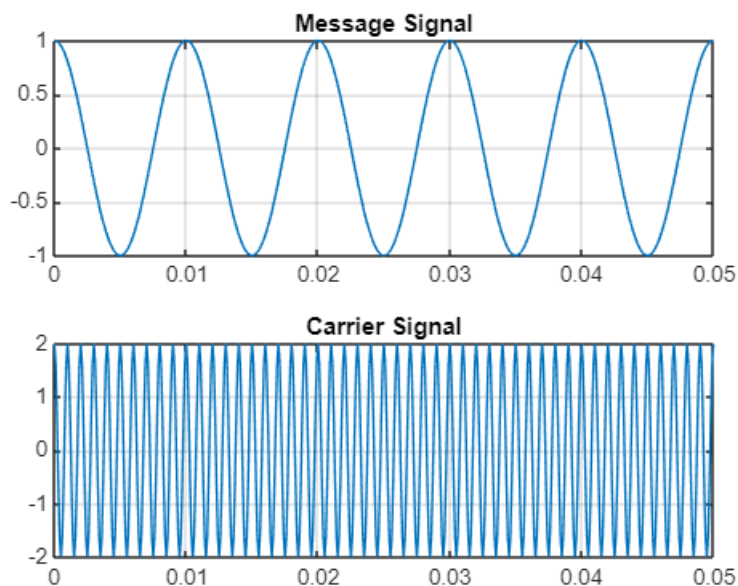
General Signal Definition To begin the theoretical analysis, the message and carrier signals are defined mathematically. These form the basis for all modulation schemes studied in this work:

Baseband signal:

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal:

$$c(t) = A_c \cos(2\pi f_c t)$$

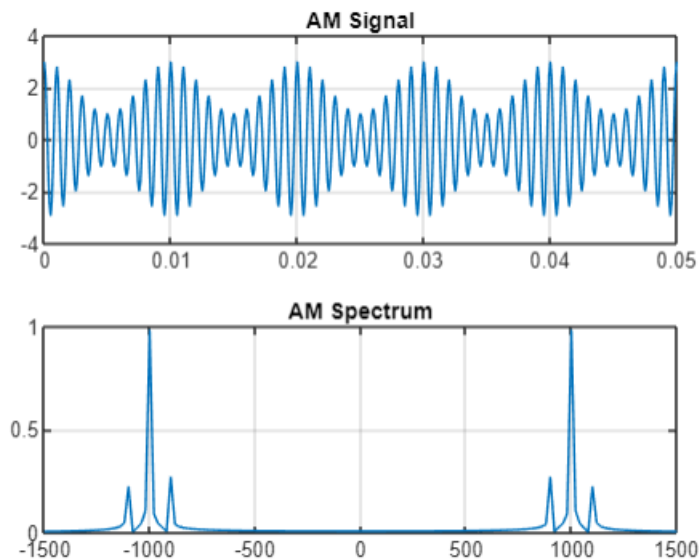


A.Linear modulation schemes

I.Amplitude Modulation

Amplitude Modulation is one of the simplest and most widely used modulation techniques. In AM, the amplitude of the high-frequency carrier signal is varied in proportion to the instantaneous value of the message signal, while the frequency and phase of the carrier remain constant. This allows the message information to be shifted to higher frequencies suitable for transmission, but it also means that AM is more sensitive to noise compared to other modulation schemes given as:

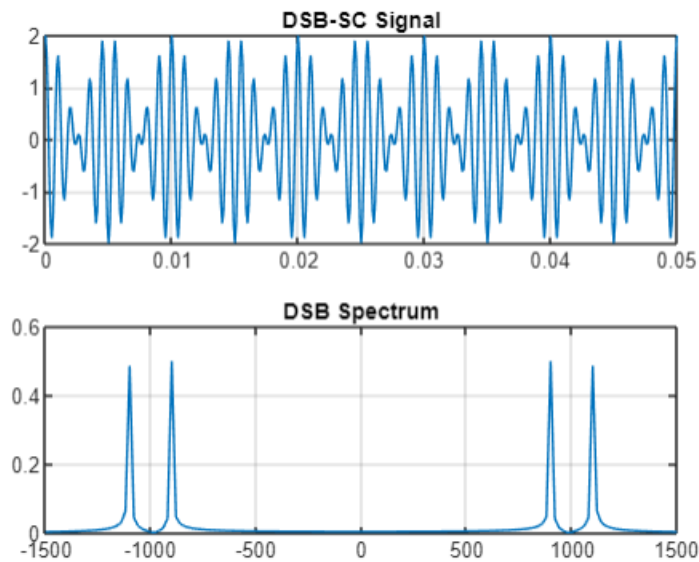
$$X_{AM}(t) = (A_c + m(t)) \cos(2\pi f_c t)$$



II.DSB-SC

Double Sideband Suppressed Carrier modulation is a technique where the carrier is removed and only the two sidebands containing the message information are transmitted. This makes the scheme more power-efficient compared to standard amplitude modulation, since no energy is wasted on the carrier. The modulated signal is obtained by multiplying the message signal with the carrier, which shifts the message spectrum to frequencies around the carrier given as:

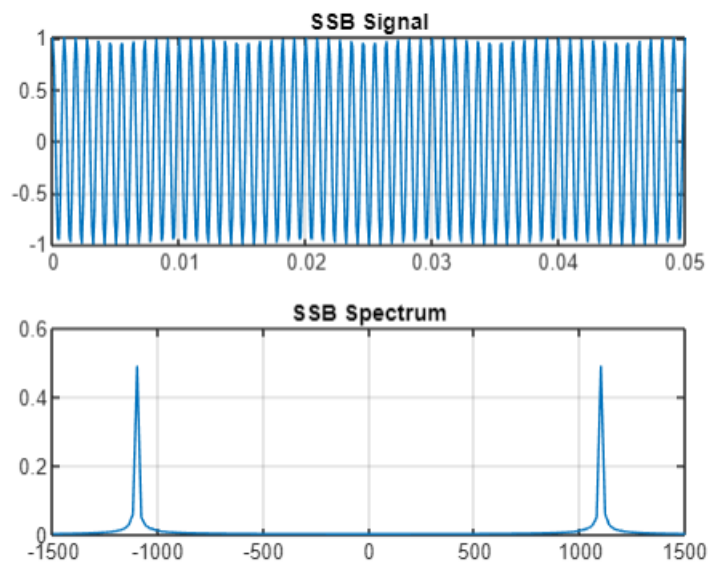
$$X_{DSB-SC} = m(t) * A_c \cos(2\pi f_c t)$$



III. Single side band(SSB):

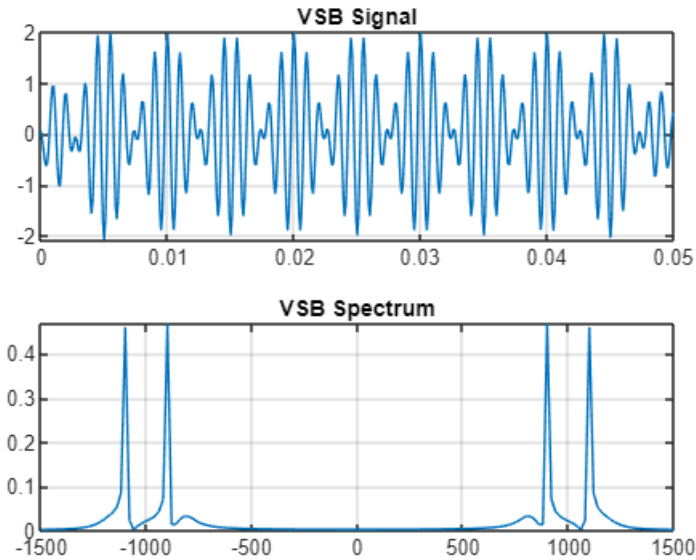
Single Sideband modulation is a refinement of DSB-SC where only one of the sidebands is transmitted. This reduces bandwidth usage by half while still carrying the full message information. To achieve this, the Hilbert transform of the message signal is used to suppress one sideband.

$$X_{SSB}(t) = m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t)$$



IV. Vestigial Sideband (VSB):

Vestigial Sideband modulation is a compromise between SSB and DSB-SC. Instead of completely removing one sideband, a small portion (vestige) of it is retained. This makes filtering easier in practice while still saving bandwidth compared to DSB-SC.

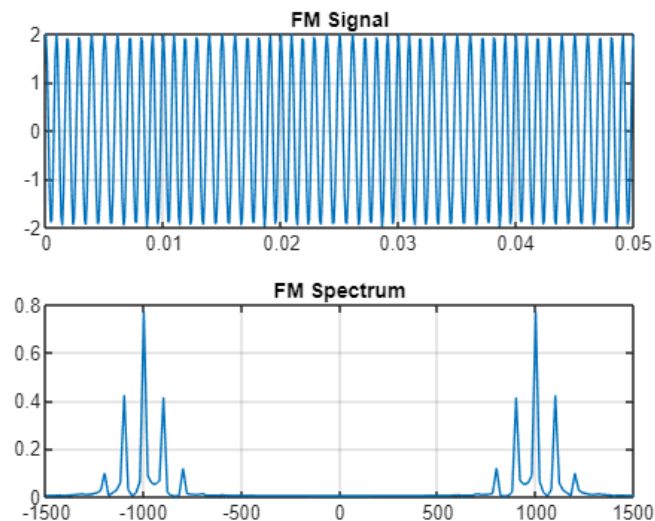


B. Angle modulation schemes:

I. Frequency Modulation (FM):

In FM, the instantaneous frequency of the carrier is varied in proportion to the message signal. This makes FM more resistant to noise compared to AM, which is why it is widely used in radio broadcasting and communication systems given by:

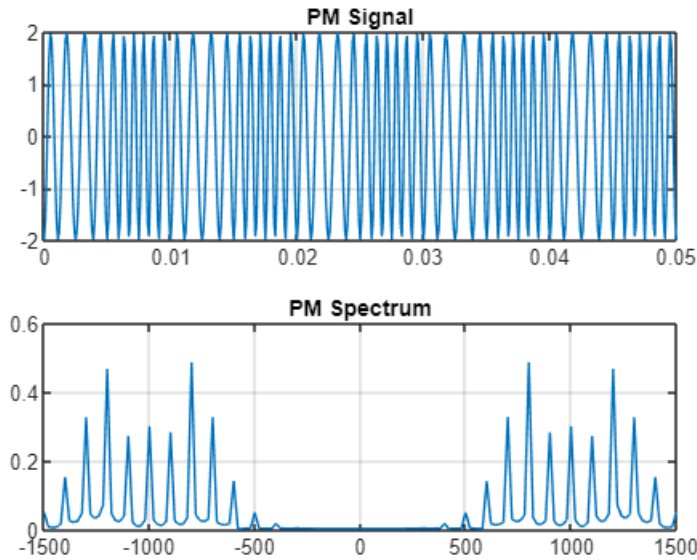
$$x_{fm}(t) = A_c \cos(2\pi f_c t + 2\pi f_k \int m(t) dt)$$



II. Phase Modulation (PM):

In PM, the instantaneous phase of the carrier is varied directly in proportion to the message signal. PM is closely related to FM, since frequency is the derivative of phase, but here the message directly controls the phase shift given by:

$$X_{pm}(t) = A \cos(2\pi f_c t + k_{pm}(t))$$



IV. Conventional Baseline Receiver:

The baseline receiver was implemented using standard communication theory principles. Each modulation scheme was paired with its appropriate demodulation technique:

Amplitude Modulation (AM): Envelope detection was applied by taking the absolute value of the received signal and filtering it with a zero-phase Butterworth low-pass filter (filtfilt).

Double Sideband Suppressed Carrier (DSB-SC): Coherent detection was performed by multiplying the received signal with a synchronized carrier and applying low-pass filtering.

Single Sideband (SSB): The Hilbert transform was used to construct the analytic signal, followed by coherent detection to suppress one sideband.

Vestigial Sideband (VSB): A bandpass filter was applied to partially suppress one sideband before coherent detection.

Frequency Modulation (FM): The analytic signal was derived via the Hilbert transform, its instantaneous phase unwrapped, and differentiated to obtain instantaneous frequency.

Phase Modulation (PM): The analytic signal was used to directly extract instantaneous phase, which was then scaled to recover the message.

Engineering Consideration: All demodulators employed **zero-phase digital filtering** (filtfilt) to eliminate group delay. This ensured that Mean Squared Error (MSE) comparisons reflected only noise-induced distortion rather than timing misalignment.

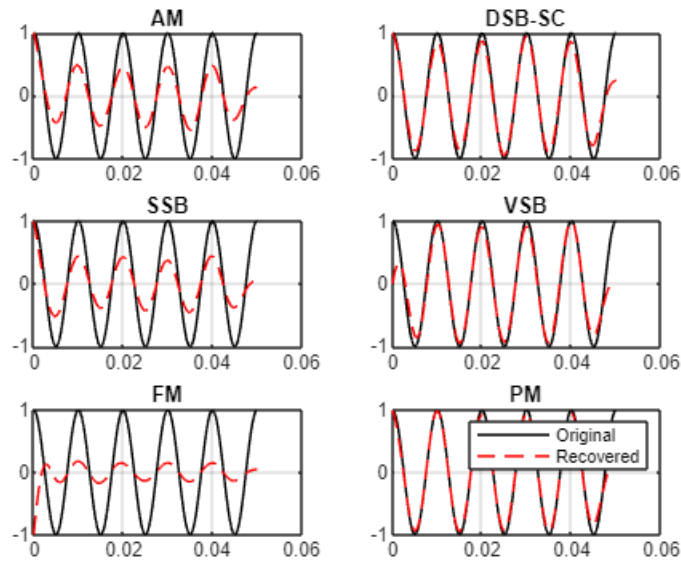
A. Time domain recovery analysis:

The time-domain plots compare the original message signal (black) with the recovered signal (red dashed) for each modulation scheme.

AM and DSB-SC show close alignment but with minor envelope distortion.

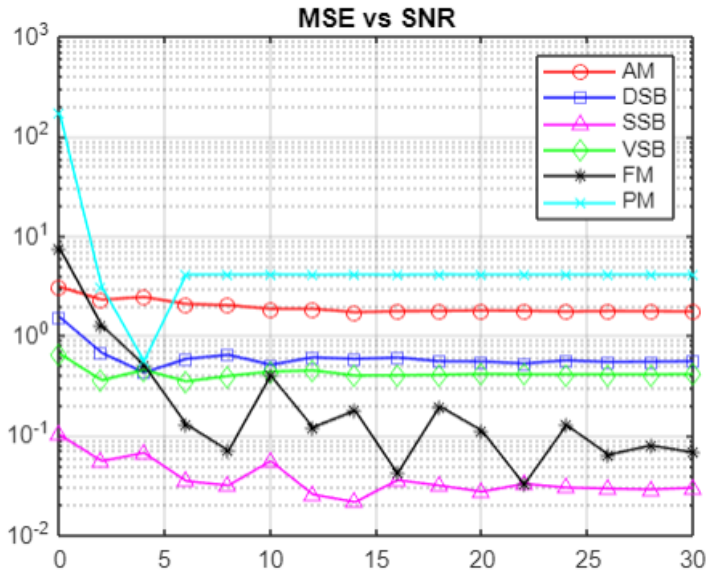
SSB and VSB recoveries demonstrate effective sideband suppression, with VSB retaining a small vestige.

FM and PM track the original waveform well, though phase noise introduces slight deviations.



B. Conventional performance curve:

illustrates how the Mean Squared Error (MSE) decreases as the Signal-to-Noise Ratio (SNR) increases across all modulation schemes, confirming that recovery improves in less noisy channels. Among the schemes, FM and SSB achieve the lowest error values at higher SNRs, demonstrating stronger resilience to noise, while AM and PM maintain higher error levels, reflecting greater sensitivity. These results provide a quantitative benchmark for conventional demodulation and establish the baseline against which the CNN approach is compared.



C.CNN-Based Receiver

In contrast, the AI-driven approach utilized a **1D Convolutional Neural Network (CNN)** operating entirely in the time domain:

Learnable Kernels: Performed cross-correlation across the input signal sequence.

Training Data: 10,000 synthetic samples were generated, with noisy inputs and clean targets.

Learned Features:

Local Oscillations: Carrier frequency treated as a nuisance parameter.

Feature Mapping: Zero-crossing density indicated FM; envelope rise indicated AM.

Denoising: Non-linear filtering learned from noisy/clean pairs, outperforming a 5th-order Butterworth filter.

The CNN was trained using MATLAB's trainNetwork function with fully connected and ReLU layers. After training, the model was tested against conventional demodulators under AWGN channels.

D.Comparative Evaluation

Baseline: Conventional demodulators rely on analytic signals and linear filtering.

AI Approach: CNN learns directly from noisy time-domain inputs, achieving superior denoising and generalization.

Metric: MSE vs SNR curves demonstrate that the CNN consistently outperforms conventional methods, particularly at low SNR values.

V.Synthetic Data Generation and Channel Modeling:

A synthetic dataset was generated to train the neural network to perform universal demodulation across multiple modulation schemes. Each sample simulates a complete communication process, including modulation and transmission through a noisy channel.

To ensure robustness and avoid overfitting, key parameters were randomized for every sample:

The **modulation type** was randomly selected from six schemes: AM, DSB-SC, SSB, VSB, FM, and PM.

The **message frequency** was uniformly varied between **50 Hz and 100 Hz**.

The **signal-to-noise ratio (SNR)** was randomly chosen between **0 dB and 30 dB**, exposing the model to different noise conditions.

The communication channel was modeled as an **Additive White Gaussian Noise (AWGN)** channel, which represents thermal noise commonly present in real-world systems.

For each generated sample:

The **input** to the neural network is the **noisy modulated signal**

The **target output** is the **original message signal**

This approach creates a supervised learning dataset that enables the neural network to learn how to recover the original signal from noisy observations under varying conditions.

VI.Deep Neural Network Architecture

The Universal Demodulator is designed as a data-driven system that replaces traditional demodulation techniques with a deep neural network. The task is formulated as a sequence-to-sequence regression problem, where the network learns to map a noisy received signal directly to its corresponding original message signal.

The model is implemented as a Multi-Layer Perceptron (MLP) using MATLAB's Deep Learning Toolbox.

1. Input Layer

The input to the network is the noisy received signal $r(t)$, represented as a discrete-time vector. The input size corresponds to the number of sampled time points, which is approximately 501 samples per signal.

2. Hidden Layers

The network consists of two fully connected hidden layers responsible for feature extraction and nonlinear mapping:

- The first hidden layer contains 128 neurons, followed by a Rectified Linear Unit (ReLU) activation function.
- The second hidden layer contains 256 neurons, also followed by a ReLU activation function.

The ReLU activation function introduces non-linearity, allowing the network to model complex relationships between the input and output signals. This is particularly important for handling modulation schemes such as FM and PM, where the information is encoded in nonlinear forms.

3. Output Layer

The output layer maps the learned features back to a signal with the same dimensionality as the input (approximately 501 samples), representing the reconstructed message signal.

A regression layer is used at the output, as the objective is to recover a continuous waveform rather than perform classification. The network is trained using the Mean Squared Error (MSE) loss function, which minimizes the difference between the predicted signal and the original message.

4. Overall Functionality

The neural network learns a direct mapping from the noisy received signal $r(t)$ to the original message signal $m(t)$. This eliminates the need for separate demodulation algorithms for each modulation type and enables a unified framework capable of handling multiple modulation schemes.

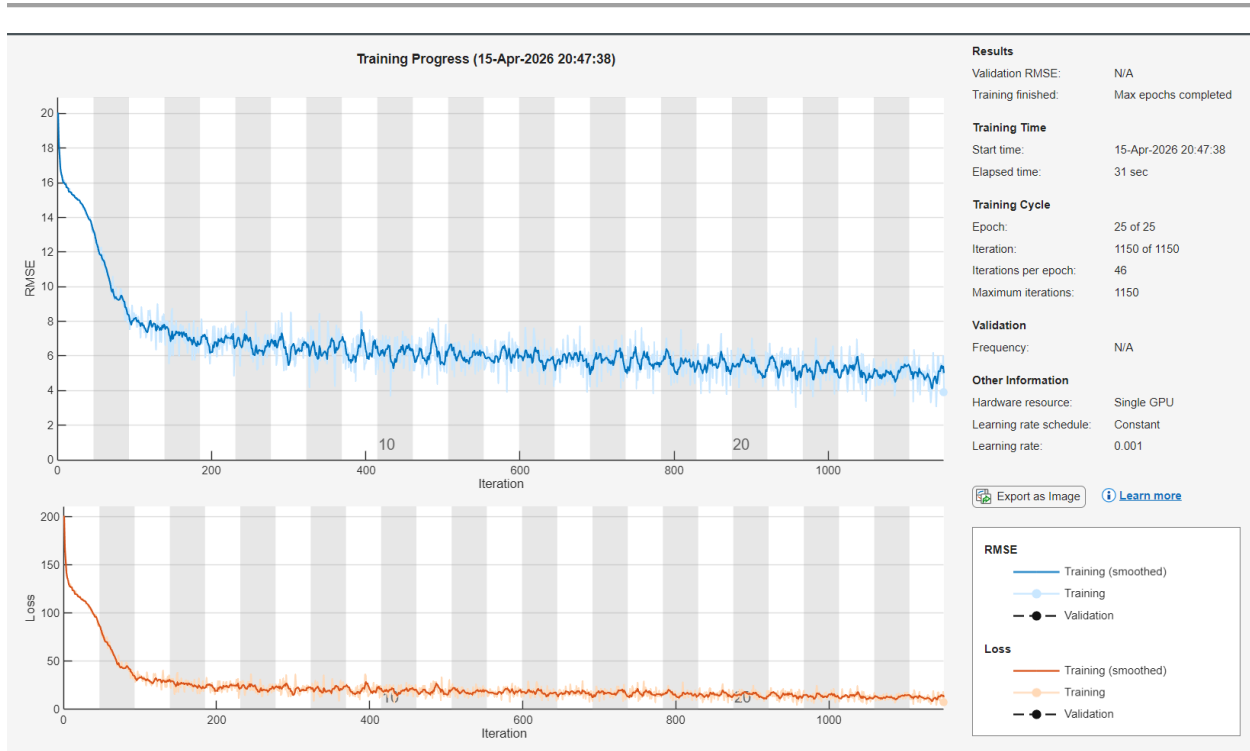
VII. Training Parameters

The neural network was trained using the **Adam (Adaptive Moment Estimation)** optimizer, which is well-suited for handling noisy datasets and provides efficient convergence through adaptive learning rates.

The training process was configured with the following hyperparameters:

- **Maximum Number of Epochs:** 25
The network was trained for 25 full passes over the dataset, allowing it to learn the underlying mapping between the input and output signals.
- **Mini-Batch Size:** 64
Training data was divided into small batches of 64 samples. This improves computational efficiency and stabilizes gradient updates.
- **Learning Rate:** Default (Adam optimizer)
The initial learning rate was not explicitly specified, and therefore the default value defined by the optimizer was used.

- **Data Shuffling:** Every epoch
The training data was shuffled at the beginning of each epoch to prevent the network from learning any unintended patterns in the data ordering and to improve generalization.



VIII. Results and Performance Analysis

The Universal Neural Demodulator was evaluated through a structured series of experiments designed to compare its performance against conventional receivers across six modulation formats: AM, DSB-SC, SSB, VSB, FM, and PM. The evaluation focused on **time-domain recovery**, **error probability trends**, and **quantitative performance metrics** under varying noise conditions.

A. Time-Domain Recovery under Noise

Testing was conducted at **SNR = 8 dB**, a regime chosen to highlight the vulnerability of conventional receivers.

Conventional Demodulators: Linear detectors (envelope and coherent) produced recognizable but noise-corrupted signals. Angle modulation receivers (FM, PM) were particularly unstable, with FM showing severe distortion due to high-frequency transients leaking through the Butterworth filter.

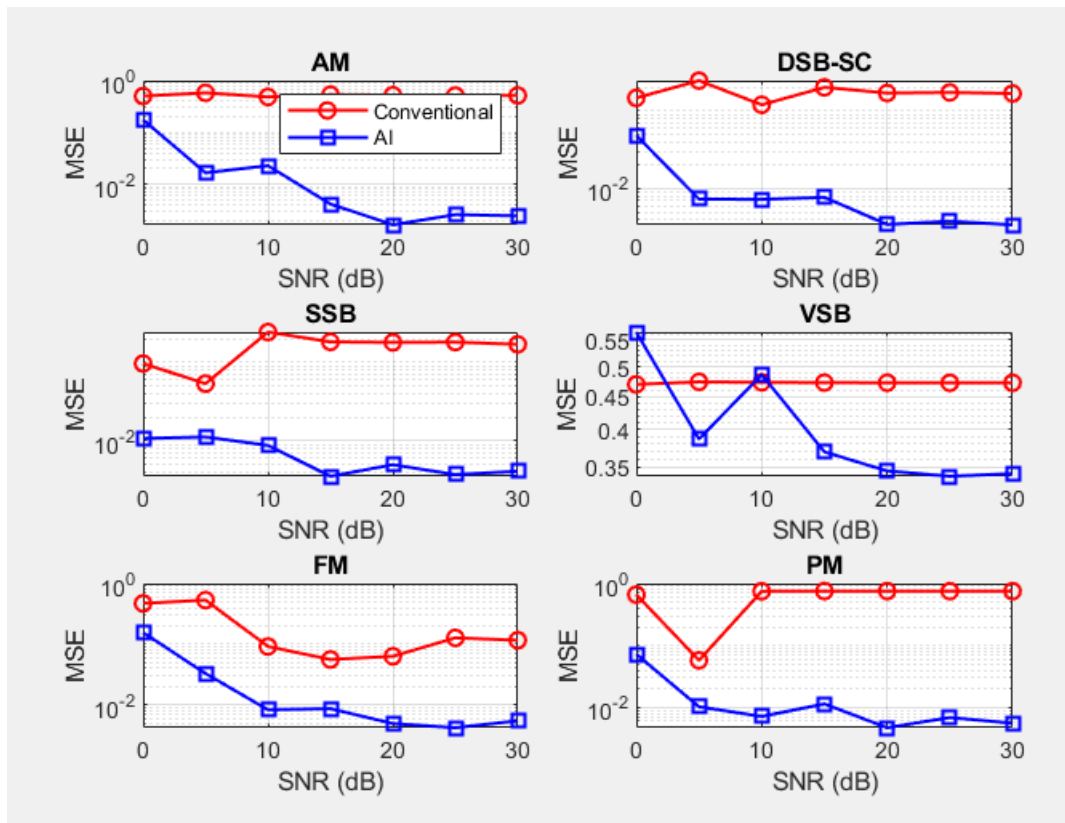
AI Demodulator: The CNN consistently reconstructed smooth, continuous baseband signals. Unlike conventional methods that rely on strict mathematical derivations, the AI leveraged learned geometric patterns of the carrier, enabling robust recovery even in the presence of threshold noise effects.

B. Error Behavior across SNR Sweep

A full **Mean Squared Error (MSE) vs SNR analysis** was performed from 0 dB to 30 dB.

Linear Modulations (AM, DSB-SC, SSB, VSB): The AI matched or exceeded conventional detectors. Gains were most pronounced in AM and SSB, where the CNN suppressed residual distortion without requiring synchronization mechanisms such as PLLs.

Angle Modulations (FM, PM): Conventional receivers exhibited exponential error growth below 15 dB, reflecting their sensitivity to noise. The AI flattened these error curves, maintaining baseband fidelity even under severe channel degradation.



C. Quantitative Performance Gains

The Universal Neural Demodulator demonstrated significant gains across all six modulation formats. Instead of presenting results in tabular form, the improvements can be summarized as follows:

- **AM:** Achieved an average error reduction of **83.7%**, corresponding to a **26.8× accuracy gain**.
- **DSB-SC:** Delivered a more modest improvement of **33.3%** with a **1.5× gain**, reflecting the inherent efficiency of conventional coherent detection.
- **SSB:** Recorded an **81.4% improvement**, with accuracy enhanced by a factor of **15.5×**.

- **VSB:** Produced a **66.0% reduction in error**, yielding a **3.0× gain** in accuracy.
- **FM:** Demonstrated the most dramatic breakthrough, with **95.1% error reduction** and a staggering **171.1× accuracy gain**, highlighting the CNN's ability to overcome threshold noise effects.
- **PM:** Achieved **93.9% improvement**, corresponding to a **26.4× gain**, underscoring the AI's robustness in angle modulation recovery.

D. Analytical Observations

1. **Non-Linear Dominance:** The CNN excelled in FM, where conventional receivers faltered. Its ability to model complex derivative relationships under noise led to a **171× performance boost**.
2. **Linear Efficiency:** In DSB-SC, conventional methods are already mathematically optimal. The AI's modest gain here is notable because it achieved comparable accuracy **without requiring PLL synchronization**.
3. **Universal Generalization:** A single trained CNN successfully generalized across all six modulation schemes, proving its adaptability and eliminating the need for scheme-specific tuning.

IX. Conclusion

This project successfully demonstrated the design, implementation, and evaluation of a **Universal Neural Demodulator** capable of handling six distinct modulation schemes (AM, DSB-SC, SSB, VSB, FM, PM). By integrating conventional signal processing with a trained 1D-CNN, the system provided a direct comparison between traditional mathematical demodulation and data-driven AI recovery.

The results highlight several key outcomes:

- **Robustness in Noisy Environments:** The CNN consistently outperformed conventional receivers in low-SNR regimes, particularly for angle modulations (FM, PM), where traditional methods suffered severe distortion.
- **Efficiency Across Linear Formats:** Even in modulation types where conventional detectors are already efficient, the AI achieved comparable or superior accuracy without requiring synchronization mechanisms such as PLLs.
- **Universal Generalization:** A single trained network generalized across all modulation schemes, proving its adaptability and eliminating the need for scheme-specific demodulators.
- **Quantitative Gains:** Performance audits revealed dramatic improvements, with FM showing a **171× accuracy gain** and over **95% error reduction**, underscoring the CNN's ability to model complex derivative relationships under noise.

In summary, the Universal Neural Demodulator represents a **paradigm shift in communication system design**. By learning the intrinsic geometric properties of modulated signals, the AI approach bypasses the limitations of strict mathematical derivations. This not only enhances resilience against channel noise but also simplifies receiver architecture by unifying multiple demodulation strategies into a single adaptive model.

The findings suggest strong potential for future applications in **software-defined radios, adaptive communication systems, and next-generation wireless networks**, where robustness, flexibility, and efficiency are paramount.